Physics Lecture Notes

June 6, 2026

Contents

Chapter 1

Introduction

1.1 Purpose of these notes

These notes were written for students of Computer Science who meet physics not as their main field of study, but as a part of their broader scientific and technical education. For this reason, the purpose of this script is not to reproduce a complete textbook of physics. It is not an encyclopedia, and it is not a catalogue of formulas. Its purpose is more modest, but also more important: to introduce selected parts of physics carefully enough that the reader can understand how physical reasoning works.

Physics is often presented to students as a collection of equations. In such a presentation, learning physics seems to mean remembering formulas and knowing when to substitute numbers into them. This is not the approach of these notes. A formula is not the beginning of understanding. A formula is a compressed statement about a physical model. To understand it, we must know what system is being described, what assumptions have been made, what quantities occur in the formula, what has been neglected, and in what range the formula can be used.

For example, an equation such as

$$F = ma$$

is not only a recipe for calculating force, mass, or acceleration. It expresses a relation between interaction and change of motion. It says something about the system, about the role of mass, about the meaning of acceleration, and about the way in which a simplified model of motion is being constructed. If the symbols are treated only as places where numbers should be inserted, then the physical content of the equation is lost.

One of the main goals of these notes is therefore to teach the reader how to read equations. Reading an equation does not mean pronouncing the symbols. It means understanding what the symbols represent, what kind of dependence is described, which quantities are variable and which are fixed, what happens when one quantity changes, what the limiting cases are, and whether the result makes physical sense. This skill is essential not only in physics, but also in mathematics, engineering, computer science, data analysis, simulation, modelling, and many other technical disciplines.

1.2 Mathematics as a tool for understanding

The reader should not expect these notes to avoid mathematics. Physics without mathematics would lose much of its power. At the same time, mathematics will be used here as a tool for understanding, not as a barrier. We will begin with mathematical preliminaries: functions, graphs, variables, vectors, derivatives, integrals, units, dimensions, and simple equations describing change. These topics will not be developed as an independent mathematics course. They will be introduced only to the extent needed for physical modelling.

The important point is that each mathematical object should carry meaning. A derivative is not only a formal operation; in kinematics it becomes velocity or acceleration, a rate of change of position or velocity. An integral is not only a method of finding an antiderivative; it can describe accumulation, displacement, work, or total effect. A vector is not only an ordered list of numbers; it represents a quantity with magnitude and direction. A graph is not decoration; it is information about behaviour. A differential equation is not only an equation with derivatives; it is often a compact way of saying how a system evolves.

1.3 From situation to model and back

The central method used throughout these notes can be summarized as a movement from a real or simplified situation to an idealized model, then to a mathematical description, then to calculation or simulation, and finally back to interpretation and checking.

We start from a situation in the real world or from a simplified physical scenario. Then we decide what is relevant and what can be neglected. We choose a system, variables, assumptions, and an idealized model. After that, we translate the model into mathematics. We calculate, analyse, or simulate. Finally, we return to physics: we interpret the result, check the units, consider limiting cases, and ask whether the answer makes sense.

This movement is not always linear. In real physical reasoning, we often move back and forth between meaning and mathematics. A calculation may suggest a physical interpretation. A physical idea may suggest a mathematical simplification. A unit check may reveal an error. A limiting case may show that the model is incomplete. A graph may help us understand an equation. A simulation may help us see what a formula predicts. For this reason, the notes will repeatedly return to the same pattern: from physical meaning to mathematical model, and from mathematical result back to physical meaning.

1.4 Why kinematics and dynamics matter

Special attention will be given to kinematics and dynamics. These are not only introductory chapters that must be completed before moving to more advanced topics. They are the backbone of the course. Kinematics introduces position, time, motion, velocity, acceleration, functions of time, and graphical descriptions of change. Dynamics introduces force, interaction, mass, inertia, Newton's laws, equations of motion, initial conditions, and prediction. In these chapters, the reader first encounters the full structure of physical modelling.

Later topics, such as oscillations, waves, energy, fields, or selected ideas from modern physics, should not be seen as disconnected areas. They reuse the same habits of thought. Again and again, we ask: What is the system? What are the relevant quantities? What assumptions are we making? What mathematical relation describes the model? What can be predicted? What does the result mean?

1.5 Depth rather than a catalogue of formulas

The notes deliberately prefer depth over the number of formulas. It is better to understand a small number of fundamental ideas well than to collect many equations without knowing how to use them. For each important topic, we will try to identify the physical situation, build the simplest useful model, state the assumptions, derive or motivate the mathematical description, and interpret the result. Some computations will be simple on purpose. Their role is not to test algebraic endurance, but to reveal structure.

This is especially important because many standard textbooks leave gaps in the reasoning. They often move quickly from a physical description to a formula, or from a formula to a result, assuming that the reader can reconstruct the intermediate steps. These notes should be more explicit. They should show not only what is done, but why it is done. They should make visible the transition from words to equations, from equations to consequences, and from consequences back to interpretation.

1.6 Units, dimensions, and checks

Units and dimensions will play an important role in this process. Checking units is one of the simplest and most powerful ways to test whether a formula or result can be correct. If a calculation gives a time measured in kilograms, something has gone wrong. Dimensional analysis is not a cosmetic detail; it is a basic tool of independent thinking.

Throughout the notes, the reader should learn to ask: What unit should the answer have? Do both sides of the equation have the same dimension? Does the result become larger or smaller in the expected way? What happens in extreme cases? Does the answer make physical sense?

1.7 Approximation and idealization

The notes will also emphasize the role of approximation and idealization. Physics does not describe reality by copying it in all its complexity. Instead, it builds models. We may neglect air resistance, treat a body as a point mass, assume a frictionless surface, use a small-angle approximation, or describe a complicated interaction with a simple force law. Such simplifications are not mistakes if they are made consciously. A model is useful when it captures the relevant structure of a situation and allows us to reason, calculate, predict, and understand. But every model has limits, and those limits must be remembered.

1.8 Computational experiments and simulations

The course will not rely on traditional laboratory experiments. However, experimental thinking will still be present. In many places, the reader may work with computational experiments or simulations. For Computer Science students this is natural and useful. A simple program or an AI-assisted HTML application can simulate a pendulum, projectile motion, oscillations, or waves. Such a simulation can allow us to change parameters, collect artificial measurements, estimate periods or frequencies, compare results with theory, and discuss measurement uncertainty.

A simulation is not reality. It is also a model. But precisely because it is a model, it can make the structure of physical reasoning visible. It allows us to see how assumptions become equations, how equations produce motion, how measurements are extracted, and how predictions are checked. Used correctly, computational experiments can help connect physics, mathematics, programming, and interpretation.

1.9 How to study these notes

The reader should approach these notes actively. Do not only look for final formulas. Ask what problem is being solved. Ask what assumptions are being made. Ask why a given quantity was chosen. Ask what the equation says before using it. Ask whether the answer has the right unit. Ask what would change if the assumptions changed. Ask whether the result is physically reasonable.

The goal of the course is not only to pass an exam. Passing the course is important, but it is not the deepest purpose of the notes. The deeper goal is to develop a way of thinking: to see physics as a disciplined movement between reality, idealization, mathematics, computation, and interpretation. A student who follows this path should become more comfortable with formulas, more careful with assumptions, more aware of the meaning of mathematical models, and more capable of using quantitative reasoning in future technical subjects.

In this sense, physics is not merely another subject to memorize. It is a training ground for modelling. It teaches how to take a complex situation, simplify it without losing what is essential, describe it mathematically, analyse the description, and return to the world with a prediction or explanation. This ability is valuable far beyond physics itself. It is one of the foundations of scientific and technical thinking.

Chapter 2

Mathematical Foundations

2.1 Why this chapter is necessary

This chapter collects the minimal mathematical language that will be used throughout these notes. It is not a replacement for a mathematics course. The reader is expected to have already met elementary algebra, functions, analytic geometry, vectors, matrices, derivatives, and integrals. However, experience shows that having passed a mathematics course does not always mean being able to use mathematics fluently in a new context.

The purpose of this chapter is therefore practical. We recall the mathematical objects and notations that will appear later, before they are used in more structured models. At this stage we do not develop applications in detail. Here the goal is to make the mathematical language readable.

Mathematics in these notes should not be treated as a collection of symbolic tricks. A formula is not only something into which numbers are substituted. A formula expresses a relation. A function describes dependence. A derivative describes local change. An integral describes accumulation. A vector describes magnitude and direction. A matrix can describe a transformation or organize several quantities at once.

The tools introduced here will return later with more specific meaning. For now, the goal is to understand their mathematical form.

2.2 Scientific notation

Physics and science in general often deal with very large and very small numbers. Writing such numbers in ordinary decimal notation is inconvenient and often hides the scale of the quantity. Scientific notation makes the scale visible.

Definition

A number is written in scientific notation if it has the form

$$a \times 10^n,$$

where n is an integer and usually $1 \leq |a| < 10$.

The number a carries the significant digits, while 10^n carries the scale. For example,

$$2.7 \times 10^{17} = 270000000000000000,$$

whereas

$$1.602 \times 10^{-19} = 0.0000000000000000001602.$$

The sign of the exponent tells us whether the decimal point is moved to the right or to the left.

This notation is not optional. It is a basic part of scientific language. Without it, many numbers used in science would be almost unreadable.

The elementary charge is approximately

$$e = 1.602 \times 10^{-19}\,\mathrm{C},$$

and the Avogadro constant is approximately

$$N_A = 6.022 \times 10^{23}\,\mathrm{mol}^{-1}.$$

The notation immediately shows that these quantities belong to completely different scales.

Scientific notation also makes multiplication and division easier. We use the rules

$$10^a \cdot 10^b = 10^{a+b}, \qquad \frac{10^a}{10^b} = 10^{a-b}.$$

Example

$$(2 \times 10^3)(3 \times 10^4) = 6 \times 10^{3+4} = 6 \times 10^7.$$

Also,

$$\frac{8 \times 10^6}{2 \times 10^2} = 4 \times 10^{6-2} = 4 \times 10^4.$$

Remark

The exponent of ten gives the order of magnitude. In many scientific situations, understanding the order of magnitude is more important than knowing many decimal digits.

The reader is expected to understand notation such as

$$10^{-6}, \qquad 10^3, \qquad 2.7 \times 10^{17}, \qquad 6.022 \times 10^{23}.$$

2.3 Numbers, quantities, and units

A number by itself is often not enough. In science, we usually work with quantities. A quantity has a numerical value, a unit, and a meaning.

For example, the expression

$$5$$

is only a number. It becomes a quantity only when we know what it represents:

$$5\,\mathrm{m}, \qquad 5\,\mathrm{s}, \qquad 5\,\mathrm{kg}, \qquad 5\,\mathrm{C}.$$

These expressions do not mean the same thing. The unit is part of the information.

Remark

Units are not decoration. They are part of the quantity and must be carried through a calculation consistently.

We can add quantities with the same unit:

$$2\,\mathrm{m} + 3\,\mathrm{m} = 5\,\mathrm{m}.$$

But the expression

$$2\,\mathrm{m} + 3\,\mathrm{s}$$

is not a meaningful sum, because metres and seconds describe different kinds of quantities.

Units also help us check formulas. If the left-hand side of an equation has a certain unit, then the right-hand side must have the same unit. This simple observation is one of the most useful ways to detect errors.

A calculation may be numerically correct and still be meaningless if the units do not match. Checking units does not solve the problem, but it protects us from many invalid results.

2.4 Variables, constants, and parameters

Mathematical notation uses symbols. These symbols may play different roles. Some symbols represent variables, some represent constants, and some represent parameters.

A variable is a quantity that is allowed to change. For example, in

$$f(x) = x^2 + 1,$$

the symbol x is the variable. Different values of x give different values of $f(x)$.

A constant has a fixed value. In

$$f(x) = 3x + 2,$$

the numbers 3 and 2 are constants.

A parameter is treated as fixed in one situation, but may have different values in different situations. The formula

$$f(x) = ax + b$$

describes a whole family of linear functions. For one particular function, a and b are fixed. If we choose different values of a and b, we get a different function.

Example

If $a = 2$ and $b = 5$, then

$$f(x) = 2x + 5.$$

If $a = -1$ and $b = 0$, then

$$f(x) = -x.$$

The formula is the same, but the parameters select different members of the family.

When reading a formula, ask which symbols change, which symbols remain fixed, and which symbols select the case being studied.

2.5 Functions

Definition

Let A and B be sets. A function from A to B is a rule that assigns to each element of A exactly one element of B.

The set A is the domain, and the set B is the codomain. If f is a function and x is an input from the domain, then

$$f(x)$$

denotes the output corresponding to x.

For example, let

$$f(x) = x^2 + 1.$$

Then

$$f(0) = 1, \qquad f(1) = 2, \qquad f(3) = 10.$$

Remark

The notation $f(x)$ should not be read as multiplication of f and x. It means: the value of the function f at the argument x.

A function may also depend on more than one variable. For example,

$$g(x, y) = x^2 + y^2$$

has two inputs. If $x = 3$ and $y = 4$, then

$$g(3, 4) = 3^2 + 4^2 = 25.$$

Functions are important not only because they allow us to compute values, but because they show how one quantity depends on another. The function

$$f(x) = 2x + 1$$

is linear: increasing x by 1 always increases the output by 2. The function

$$f(x) = x^2$$

is nonlinear: the change in the output depends on the current value of x.

2.6 Cartesian coordinates

To describe points on a line, in a plane, or in space, we use coordinates. The most common coordinate system is the Cartesian coordinate system.

Setting	Coordinates	Example
Line	x	2
Plane	(x, y)	$(2, 3)$
Space	(x, y, z)	$(1, -2, 5)$

This horizontal summary is often more useful than writing the three cases separately. It shows at once how the number of coordinates grows with the dimension.

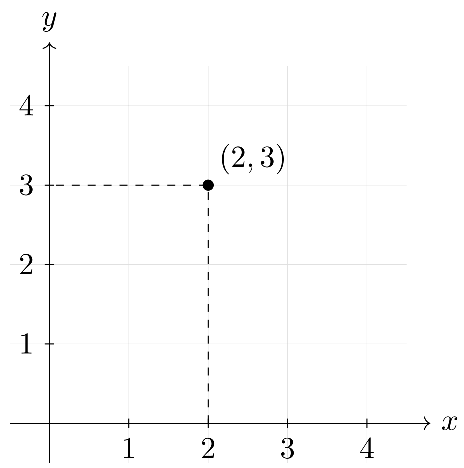

Figure 2.1: A point represented by Cartesian coordinates in the plane.

Coordinates are naturally connected with the way we describe location. A coordinate system gives a mathematical way of saying where something is.

Remark

Coordinates are not the point itself. Coordinates depend on a choice of origin, axes, orientation, and units. The same point can have different coordinates in different coordinate systems.

Geometry concerns objects such as points, lines, planes, and curves. Coordinates are a method of describing these objects using numbers.

2.7 Graphs

A graph is a visual representation of a relation. For a function

$$y = f(x),$$

we usually place x on the horizontal axis and y on the vertical axis. Each point on the graph has coordinates

$$(x, f(x)).$$

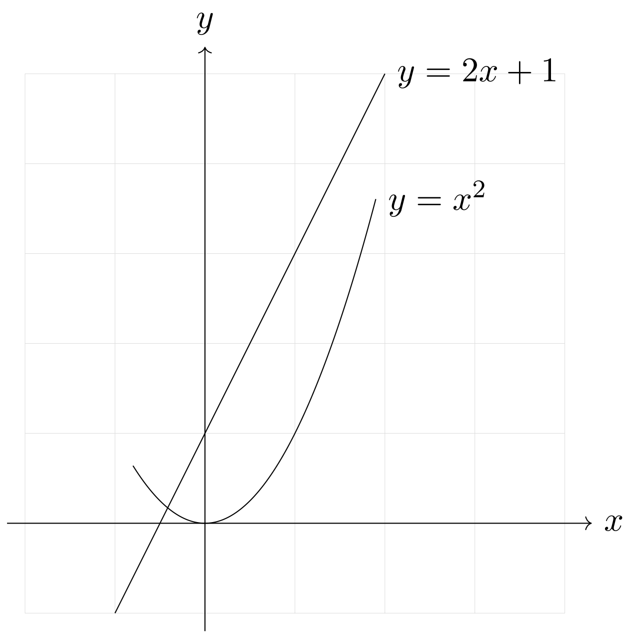

Figure 2.2: Graphs of a linear function and a quadratic function.

A graph is not only a picture. It contains information. From a graph we can often see where the function increases or decreases, where it changes rapidly or slowly, where it has zeros, and whether its shape is linear or nonlinear.

For example, the graph of $f(x) = 2x + 1$ is a straight line. The graph of $f(x) = x^2$ is a parabola.

2.8 Curves and parametric descriptions

A curve can be described in different ways. One way is to give an equation such as

$$y = f(x).$$

For example, $y = x^2$ describes a parabola in the plane.

Another way is to use a parameter. A parameter is an additional variable that describes how the coordinates are generated.

Definition

A parametric curve in the plane is a mapping

$$\mathbf{r}\colon I \to \mathbb{R}^2, \qquad \mathbf{r}(t) = (x(t), y(t)),$$

where $I \subseteq \mathbb{R}$ is an interval and $x, y\colon I \to \mathbb{R}$ are coordinate functions.

As t varies in I, the point $(x(t), y(t))$ traces the curve.

For example, the equations

$$x = t, \qquad y = t^2$$

give the same parabola as $y = x^2$, because the moving point is

$$(x, y) = (t, t^2).$$

Another important example is

$$\mathbf{r}(t) = (\cos t, \sin t).$$

This describes the unit circle, because

$$\cos^2 t + \sin^2 t = 1.$$

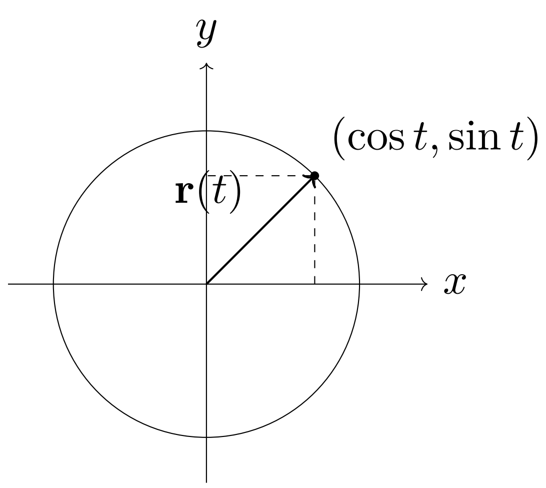

Figure 2.3: Parametric description of the unit circle.

Remark

At this stage, t is only a mathematical parameter. The important idea is that the coordinates of a moving point are functions of one parameter.

2.9 Vectors

A vector is a mathematical object with magnitude and direction. In coordinates, a vector in the plane may be written as

$$\mathbf{v} = (v_x, v_y),$$

and a vector in space as

$$\mathbf{v} = (v_x, v_y, v_z).$$

The numbers v_x, v_y, and v_z are called components. They depend on the coordinate system.

Example

The vector

$$\mathbf{v} = (3, 4)$$

has magnitude

$$|\mathbf{v}| = \sqrt{3^2 + 4^2} = 5.$$

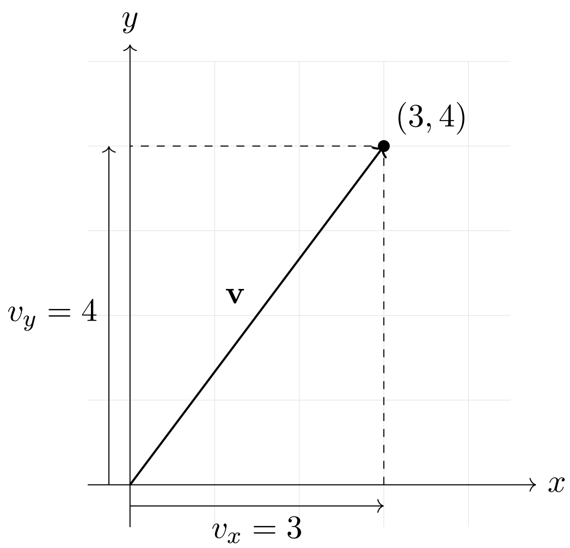

Figure 2.4: Components of a vector in the plane.

Vectors can be added component by component:

$$(1, 2) + (3, 4) = (4, 6).$$

They can be multiplied by a scalar:

$$2(1, 3) = (2, 6).$$

The zero vector is $\mathbf{0} = (0, 0)$ in the plane and $\mathbf{0} = (0, 0, 0)$ in space.

The dot product of two vectors is defined by

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y$$

in two dimensions, and by

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$$

in three dimensions.

The dot product is also related to the angle between vectors:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta.$$

This formula shows that the dot product contains information about both lengths and direction.

2.10 Matrices

Definition

A matrix is a rectangular array of numbers arranged in rows and columns.

For example,

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Matrices can be added when they have the same size:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}.$$

A matrix can be multiplied by a scalar:

$$2\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}.$$

A matrix can also multiply a vector:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + 2y \\ 3x + 4y \end{pmatrix}.$$

Matrix multiplication is not the same as ordinary multiplication of numbers. In general,

$$AB \neq BA.$$

This means that the order of multiplication matters.

Matrices are useful because they can represent linear transformations. For example, the matrix

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

represents a rotation in the plane by an angle θ. If

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix},$$

then $R(\theta)\mathbf{x}$ is the rotated vector.

Remark

In these notes, matrices will not be developed in detail. The reader should remember the basic idea: matrices organize several numbers and can describe transformations acting on vectors or systems of quantities.

2.11 Derivatives

The derivative describes how a function changes locally. If

$$y = f(x),$$

then the derivative of f with respect to x is written as

$$f'(x) \qquad \text{or} \qquad \frac{df}{dx}.$$

Definition

Let f be a real-valued function defined near a. If the limit

$$\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$$

exists, then it is called the derivative of f at a and is denoted by $f'(a)$.

For example, if

$$f(x) = x^2,$$

then

$$f'(x) = 2x.$$

At $x = 1$, we get $f'(1) = 2$. At $x = 3$, we get $f'(3) = 6$. The function changes more rapidly near $x = 3$ than near $x = 1$.

If

$$f(x) = 3x + 2,$$

then

$$f'(x) = 3.$$

The derivative is constant because the function is linear and has the same slope everywhere. If $f(x) = C$, where C is constant, then $f'(x) = 0$.

The derivative also has a geometric interpretation. It is the slope of the tangent line to the graph of the function. If the derivative is positive, the function is increasing. If the derivative is negative, the function is decreasing. If the derivative is zero, the function is locally flat.

Some basic differentiation rules are

$$\frac{d}{dx}x^n = nx^{n-1}, \qquad \frac{d}{dx}(f(x)+g(x)) = f'(x)+g'(x), \qquad \frac{d}{dx}(cf(x)) = cf'(x),$$

where c is a constant.

Example

$$\frac{d}{dx}(3x^2 + 5x - 7) = 6x + 5.$$

In this chapter, the important point is purely mathematical: a derivative measures local change.

2.12 Integrals

An integral describes accumulation. It can be understood as a limiting sum or, geometrically, as the signed area under a graph. In this introductory chapter we mainly need the connection between integration and antiderivatives.

Definition

Let $f\colon I \to \mathbb{R}$ be a function defined on an interval I. A function $F\colon I \to \mathbb{R}$ is called an antiderivative of f on I if F is differentiable on I and

$$F'(x) = f(x)$$

for every $x \in I$.

Definition

The indefinite integral of f is the family of all antiderivatives of f. If F is one antiderivative of f, we write

$$\int f(x)\,dx = F(x) + C,$$

where C is an arbitrary constant.

For example,

$$\int 2x\,dx = x^2 + C,$$

because

$$\frac{d}{dx}(x^2 + C) = 2x.$$

Remark

The constant C appears because the derivative of a constant is zero. Therefore many functions can have the same derivative.

The definite integral of $f(x)$ from a to b is written as

$$\int_a^b f(x)\,dx.$$

It measures accumulation over the interval $[a, b]$. Its precise construction can be given using limiting sums; in these notes we will mainly compute it using the theorem below.

Theorem

If f is continuous on $[a, b]$ and F is an antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Example

Since $F(x) = x^2$ is an antiderivative of $f(x) = 2x$, we have

$$\int_0^3 2x\,dx = \left[x^2\right]_0^3 = 9 - 0 = 9.$$

The symbol dx indicates the variable of integration. In the expression

$$\int_a^b f(x)\,dx,$$

we integrate with respect to x.

Some basic integration rules are

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1),$$

$$\int (f(x) + g(x))\,dx = \int f(x)\,dx + \int g(x)\,dx, \qquad \int cf(x)\,dx = c\int f(x)\,dx.$$

Example

$$\int (3x^2 + 5x - 7)\, dx = x^3 + \frac{5}{2}x^2 - 7x + C.$$

In this chapter, the important mathematical idea is that integration describes accumulation and is closely connected with differentiation.

2.13 Differentiation and integration as inverse operations

Differentiation and integration are closely related. In a basic sense, they are inverse operations.

If

$$F'(x) = f(x),$$

then

$$\int f(x)\, dx = F(x) + C.$$

Example

Since

$$\frac{d}{dx}x^3 = 3x^2,$$

we have

$$\int 3x^2\, dx = x^3 + C.$$

For definite integrals, the connection is given by the fundamental theorem of calculus.

Theorem

If f is continuous on $[a, b]$ and F is an antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\, dx = F(b) - F(a).$$

Example

If $f(x) = 2x$, then we can choose $F(x) = x^2$. Thus

$$\int_1^4 2x\, dx = F(4) - F(1) = 4^2 - 1^2 = 15.$$

2.14 Simple differential equations

Definition

A differential equation is an equation involving an unknown function and one or more of its derivatives.

Such an equation describes a relation between a function and its change. Unlike an algebraic equation, it does not usually ask for a single number. It asks for a function.

The equation

$$\frac{dy}{dx} = 3$$

asks for a function whose derivative is 3. The general solution is

$$y(x) = 3x + C.$$

Another simple example is

$$\frac{dy}{dx} = 2x.$$

A solution is

$$y(x) = x^2 + C,$$

because

$$\frac{d}{dx}(x^2 + C) = 2x.$$

A slightly different example is

$$\frac{dy}{dx} = ky,$$

where k is a constant. This equation says that the derivative of the function is proportional to the function itself. One family of solutions is

$$y(x) = Ce^{kx}.$$

Indeed,

$$\frac{d}{dx}Ce^{kx} = kCe^{kx} = ky(x).$$

The constant C is determined by an additional condition. For example, if

$$y(0) = 2,$$

then

$$2 = Ce^{k\cdot 0} = C,$$

so

$$y(x) = 2e^{kx}.$$

Remark

Here the important point is the basic structure: a differential equation contains derivatives and asks for an unknown function.

2.15 Approximation and limiting cases

Mathematics often uses exact expressions, but in applications we frequently use approximations. An approximation replaces a complicated expression with a simpler one under specified conditions.

For small x measured in radians,

$$\sin x \approx x.$$

Also, for small x,

$$(1 + x)^n \approx 1 + nx.$$

Remark

An approximation is useful only together with its condition of validity. Without that condition, the approximation is incomplete information.

A limiting case is a way of checking an expression by asking what happens when a variable becomes very small, very large, or takes a special value.

For example, consider

$$f(x) = \frac{x}{1+x}.$$

If x is very small, then

$$f(x) \approx x.$$

If x is very large, then

$$f(x) \approx 1.$$

These limiting behaviours help us understand the structure of the function.

Chapter 3

Kinematics

3.1 What kinematics studies

Definition

Kinematics is the part of mechanics that describes motion without explaining its causes.

At this stage we do not introduce forces, interactions, or causes of motion. Those questions belong to dynamics. In kinematics we ask a more basic question:

How can motion be described mathematically?

To answer this question, we use the mathematical language recalled in the previous chapter: coordinates, functions, parametric curves, vectors, derivatives, and integrals. The new idea is not a new mathematical object. The new idea is interpretation. We now use mathematical objects to describe the motion of a body in space and time.

The central object of kinematics is the position of a moving point as a function of time. Once position is known, we can ask how it changes. This leads to velocity. Once velocity is known, we can ask how velocity changes. This leads to acceleration:

$$\text{position} \quad \longrightarrow \quad \text{velocity} \quad \longrightarrow \quad \text{acceleration.}$$

Mathematically, this direction is built using derivatives. Later we will also move in the opposite direction:

$$\text{acceleration} \quad \longrightarrow \quad \text{velocity} \quad \longrightarrow \quad \text{position.}$$

This opposite direction is built using integrals and initial conditions.

Remark

Kinematics describes motion. Dynamics explains why motion changes. Keeping these two questions separate is essential at the beginning.

This chapter should be read slowly. The goal is not only to remember formulas, but to understand how a mathematical description of motion is constructed.

3.2 Position as a function of time

3.2.1 Position in a coordinate system

To describe motion, we must first describe position. A position is described only after we choose a coordinate system. Coordinates are therefore not the point itself; they are a numerical description of the point relative to a chosen origin, axes, orientation, and units.

Definition

In a chosen coordinate system, the position vector of a point is the vector from the origin of the coordinate system to that point.

In one dimension, position is represented by one coordinate x. In two dimensions, position may be written as (x, y), and in three dimensions as (x, y, z). It is often convenient to write position as a vector:

$$\mathbf{r} = (x, y) \qquad \text{or} \qquad \mathbf{r} = (x, y, z).$$

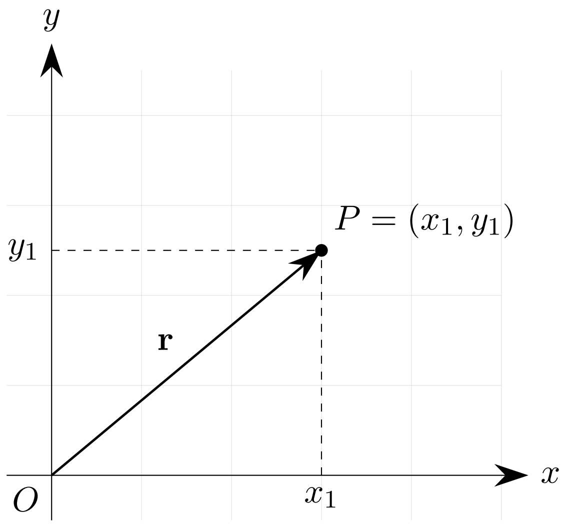

Figure 3.1: Position vector in a two-dimensional coordinate system.

Remark

If we choose a different origin or different axes, the coordinates of the same physical point may change. This is not a contradiction: coordinates depend on the chosen coordinate system.

3.2.2 Motion as changing position

A body is in motion when its position changes in time. This means that the coordinates of the point are functions of time.

Definition

A motion of a point in space is described by a position function

$$\mathbf{r} \colon I \to \mathbb{R}^3, \qquad \mathbf{r}(t) = (x(t), y(t), z(t)),$$

where $I \subseteq \mathbb{R}$ is an interval of time.

In one dimension we write $x = x(t)$. In two dimensions we write

$$\mathbf{r}(t) = (x(t), y(t)),$$

and in three dimensions

$$\mathbf{r}(t) = (x(t), y(t), z(t)).$$

The expression $\mathbf{r}(t)$ means that position depends on time. Choosing a value of t gives the position of the body at that time.

Example

If

$$\mathbf{r}(t) = (t, t^2),$$

then

$$\mathbf{r}(0) = (0,0), \qquad \mathbf{r}(1) = (1,1), \qquad \mathbf{r}(2) = (2,4).$$

One formula describes many positions: it tells us where the point is at each time.

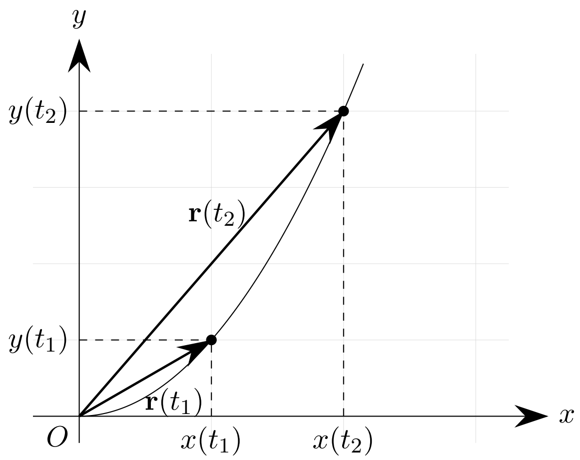

Figure 3.2: Position vectors at two different times for the motion $\mathbf{r}(t) = (t, t^2)$.

The curve shows the possible positions of the point. The labels t_1 and t_2 mark two different moments of time. At each moment, the point has a different position vector.

3.2.3 Trajectory

Definition

The trajectory of a moving point is the set of all positions occupied by that point during the motion:

$$\{\mathbf{r}(t) : t \in I\}.$$

The trajectory is a geometric curve. The motion contains more information, because it tells us where the point is at each moment of time. Two different motions may therefore have the same trajectory.

Example

The motions

$$\mathbf{r}_1(t) = (\cos t, \sin t) \qquad \text{and} \qquad \mathbf{r}_2(t) = (\cos 2t, \sin 2t)$$

both lie on the unit circle, because in both cases $x^2 + y^2 = 1$. The trajectory is the same, but the time description is different.

Geometry alone tells us the shape of the path. Kinematics needs more: it needs the position as a function of time.

3.2.4 Simple examples of position functions

Example

For motion along a straight line, one may write

$$x(t) = x_0 + at,$$

where x_0 and a are constants. If $x_0 = 1$ and $a = 2$, then

$$x(t) = 1 + 2t, \qquad x(0) = 1, \qquad x(1) = 3, \qquad x(2) = 5.$$

The formula gives the position at every time.

In two dimensions, a straight-line motion can be written as

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{a},$$

where $\mathbf{r}_0 = (x_0, y_0)$ and $\mathbf{a} = (a_x, a_y)$. In coordinates,

$$x(t) = x_0 + a_x t, \qquad y(t) = y_0 + a_y t.$$

Example

Consider

$$\mathbf{r}(t) = (t, t^2).$$

Then $x(t) = t$ and $y(t) = t^2$. Eliminating t, we get $t = x$, hence

$$y = x^2.$$

The trajectory is a parabola, while $\mathbf{r}(t) = (t, t^2)$ also contains the time description of motion along that parabola.

Another parametrization may give the same parabola with a different time description. For example,

$$\mathbf{r}(t) = (2t, 4t^2)$$

also satisfies $y = x^2$, because if $x = 2t$, then $t = x/2$, and

$$y = 4t^2 = 4\left(\frac{x}{2}\right)^2 = x^2.$$

Example

For circular motion, consider

$$\mathbf{r}(t) = (R\cos t, R\sin t),$$

where $R > 0$ is constant. Then

$$x^2(t) + y^2(t) = R^2\cos^2 t + R^2\sin^2 t = R^2.$$

Thus the trajectory is a circle of radius R centered at the origin.

The equation

$$x^2 + y^2 = R^2$$

describes the circle as a geometric object. The parametric formula $\mathbf{r}(t) = (R\cos t, R\sin t)$ describes the position on the circle at each time.

We may also write

$$\mathbf{r}(t) = (R\cos(\omega t), R\sin(\omega t)),$$

where ω is a constant. This describes the same circle, but the time dependence is different. We will return to the meaning of ω when we discuss velocity and circular motion.

3.2.5 What position tells us and what it does not tell us

A function $\mathbf{r}(t)$ is the complete kinematic description of position. It tells us where the point is at every time. From it, we can find the trajectory by looking at the set of all points $\mathbf{r}(t)$.

However, position alone is not the end of the description. Once we know how position depends on time, we naturally ask how fast the position changes and in what direction it changes. This question leads to velocity.

3.3 Velocity

3.3.1 Why position is not enough

Knowing the position of a moving point at every time tells us where the point is. But we often want a more local description: how the position is changing at a given moment.

If the position changes a lot during a short time interval, we say that the motion is fast. If the position changes only a little during the same time interval, we say that the motion is slow. If the position changes in a different direction, the motion has a different direction.

Velocity is the mathematical object that describes this change of position.

3.3.2 Average velocity

Let the position of a moving point be $\mathbf{r}(t)$. Suppose we compare two moments of time, t_1 and t_2, with $t_1 \neq t_2$. The displacement over this interval is

$$\Delta\mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1),$$

and the elapsed time is

$$\Delta t = t_2 - t_1.$$

Definition

The average velocity of a moving point on the interval $[t_1, t_2]$, with $t_1 \neq t_2$, is

$$\mathbf{v}_{\text{avg}} = \frac{\Delta\mathbf{r}}{\Delta t} = \frac{\mathbf{r}(t_2) - \mathbf{r}(t_1)}{t_2 - t_1}.$$

The average velocity is a vector. Its direction is the direction of the displacement $\Delta\mathbf{r}$, and its magnitude tells us how much displacement occurs per unit time during the interval. In one dimension,

$$v_{\text{avg}} = \frac{x(t_2) - x(t_1)}{t_2 - t_1}.$$

Remark

Average velocity depends on the interval. A large interval describes overall change; a smaller interval gives a more local description.

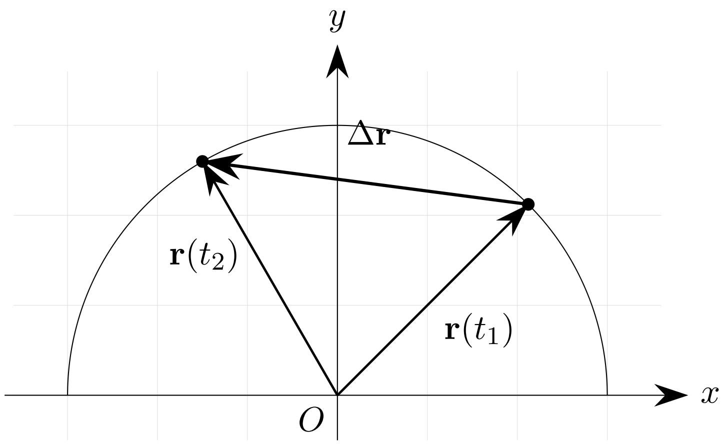

Figure 3.3: Displacement between two positions of a moving point.

3.3.3 Instantaneous velocity

Average velocity describes motion over a time interval. To describe motion at one instant, we shrink the interval.

Definition

Let $\mathbf{r}$ be differentiable at time t. The instantaneous velocity at time t is

$$\mathbf{v}(t) = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{d\mathbf{r}}{dt}(t).$$

The limiting process is essential. We are not dividing by zero. We first compute the average velocity over a nonzero time interval Δt, and only then study what this expression approaches as Δt becomes arbitrarily small.

If

$$\mathbf{r}(t) = (x(t), y(t), z(t)),$$

then

$$\mathbf{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right).$$

In two dimensions,

$$\mathbf{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}\right),$$

and in one dimension,

$$v(t) = \frac{dx}{dt}.$$

Velocity is therefore the derivative of position with respect to time.

3.3.4 Velocity as a vector attached to the moving point

Velocity is a vector quantity. It has magnitude and direction. At each time t, the point has a position $\mathbf{r}(t)$ and a velocity $\mathbf{v}(t)$. It is useful to imagine the velocity vector as attached to the point at its current position.

The velocity vector is tangent to the trajectory. The trajectory tells us the path. The velocity tells us the local direction in which the point is moving along that path.

Remark

For motion along a circle, velocity is tangent to the circle, not directed toward the center. The direction of velocity changes from point to point, even if the speed remains constant.

3.3.5 Speed

Definition

The speed of a moving point at time t is the magnitude of its velocity vector:

$$v(t) = |\mathbf{v}(t)|.$$

If

$$\mathbf{v}(t) = (v_x(t), v_y(t), v_z(t)),$$

then

$$|\mathbf{v}(t)| = \sqrt{v_x^2(t) + v_y^2(t) + v_z^2(t)}.$$

Speed is not a vector. It is a non-negative number. It tells us how fast the point is moving, but not in which direction. Velocity tells us both: how fast and in which direction.

3.3.6 Examples

Example

For one-dimensional motion

$$x(t) = x_0 + at,$$

where x_0 and a are constants, the velocity is

$$v(t) = \frac{dx}{dt} = a.$$

If $x(t) = 1 + 2t$, then $v(t) = 2$. The number 2 is the rate at which position changes with respect to time.

Example

For

$$\mathbf{r}(t) = (t, t^2),$$

we have

$$\mathbf{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}\right) = (1, 2t).$$

Thus

$$\mathbf{v}(0) = (1, 0), \qquad \mathbf{v}(1) = (1, 2), \qquad \mathbf{v}(2) = (1, 4).$$

The horizontal component is constant, while the vertical component changes.

Example

For circular motion

$$\mathbf{r}(t) = (R\cos t, R\sin t),$$

the velocity is

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = (-R\sin t, R\cos t).$$

At $R = 2$ and $t = \pi/3$,

$$\mathbf{r}\left(\frac{\pi}{3}\right) = (1, \sqrt{3}), \qquad \mathbf{v}\left(\frac{\pi}{3}\right) = (-\sqrt{3}, 1).$$

Their dot product is

$$(1, \sqrt{3}) \cdot (-\sqrt{3}, 1) = 0,$$

so at this point the velocity vector is perpendicular to the position vector.

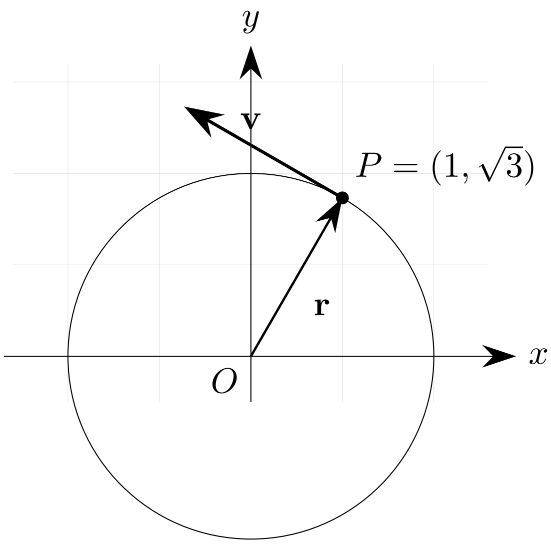

Figure 3.4: Velocity tangent to a circular trajectory.

The speed in this circular motion is

$$|\mathbf{v}(t)| = \sqrt{(-R\sin t)^2 + (R\cos t)^2} = R.$$

The speed is constant, although the velocity vector changes direction.

3.4 Acceleration

3.4.1 Why velocity is not the end of the description

Velocity describes how position changes. But velocity itself may also change. A body may move faster and faster, slow down, or move along a curve while its direction of motion changes.

Acceleration describes the change of velocity with respect to time.

3.4.2 Average acceleration

Suppose the velocity at time t_1 is $\mathbf{v}(t_1)$, and the velocity at time t_2 is $\mathbf{v}(t_2)$, with $t_1 \neq t_2$. The change in velocity is

$$\Delta\mathbf{v} = \mathbf{v}(t_2) - \mathbf{v}(t_1).$$

Definition

The average acceleration of a moving point on the interval $[t_1, t_2]$, with $t_1 \neq t_2$, is

$$\mathbf{a}_{\text{avg}} = \frac{\Delta \mathbf{v}}{\Delta t} = \frac{\mathbf{v}(t_2) - \mathbf{v}(t_1)}{t_2 - t_1}.$$

Average acceleration describes how velocity changes over a time interval.

3.4.3 Instantaneous acceleration

Definition

Let $\mathbf{v}$ be differentiable at time t. The instantaneous acceleration at time t is

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt}(t).$$

Since velocity is the derivative of position,

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt},$$

we also have

$$\mathbf{a}(t) = \frac{d^2\mathbf{r}}{dt^2}.$$

Thus acceleration is the second derivative of position with respect to time.

In coordinates,

$$\mathbf{a}(t) = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}, \frac{d^2z}{dt^2}\right).$$

In two dimensions,

$$\mathbf{a}(t) = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}\right),$$

and in one dimension,

$$a(t) = \frac{d^2x}{dt^2}.$$

3.4.4 Acceleration is also a vector

Acceleration is a vector. It has magnitude and direction. It describes how the velocity vector changes.

Since velocity is a vector, it can change in magnitude, in direction, or in both. Therefore a body may be accelerating even if its speed is constant, provided that the direction of velocity changes.

Remark

Constant speed does not necessarily mean zero acceleration. If the direction of velocity changes, the velocity vector changes, and the acceleration is not zero.

3.4.5 Examples

Example

Consider one-dimensional motion

$$x(t) = x_0 + at,$$

where x_0 and a are constants. Then

$$v(t) = a.$$

The acceleration is

$$a_{\mathrm{kin}}(t) = \frac{dv}{dt} = 0.$$

Here a_{kin} denotes acceleration, while a is a parameter in the formula for $x(t)$.

Example

For

$$\mathbf{r}(t) = (t, t^2),$$

we found

$$\mathbf{v}(t) = (1, 2t).$$

Therefore

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = (0, 2).$$

The acceleration is constant and points in the positive y-direction.

Example

For circular motion

$$\mathbf{r}(t) = (R\cos t, R\sin t),$$

we found

$$\mathbf{v}(t) = (-R\sin t, R\cos t).$$

Differentiating again gives

$$\mathbf{a}(t) = (-R\cos t, -R\sin t).$$

Since

$$\mathbf{r}(t) = (R\cos t, R\sin t),$$

we get

$$\mathbf{a}(t) = -\mathbf{r}(t).$$

The acceleration points toward the center of the circle.

This is a major conceptual point. In uniform circular motion, speed may be constant, but acceleration is not zero, because the direction of velocity is changing.

3.4.6 Units of acceleration

Acceleration has units of velocity divided by time. If velocity is measured in metres per second and time in seconds, acceleration is measured in

$$\mathrm{m/s^2}.$$

This also follows from

$$\mathbf{a}(t) = \frac{d^2\mathbf{r}}{dt^2}.$$

Position contributes metres, and two differentiations with respect to time contribute two factors of seconds in the denominator. The unit $\mathrm{m/s^2}$ means metres per second per second.

3.5 Moving between position, velocity, and acceleration

3.5.1 The derivative direction

So far we have moved in one direction:

$$\mathbf{r}(t) \quad \longrightarrow \quad \mathbf{v}(t) \quad \longrightarrow \quad \mathbf{a}(t).$$

This direction uses derivatives:

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}, \qquad \mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}.$$

If we know position as a function of time, we can compute velocity and acceleration by differentiating. No additional information is needed for this direction. The function $\mathbf{r}(t)$ already contains all information about the motion.

Example

If

$$\mathbf{r}(t) = (t, t^2),$$

then

$$\mathbf{v}(t) = (1, 2t), \qquad \mathbf{a}(t) = (0, 2).$$

Everything follows by differentiation.

3.5.2 The integral direction

We can also move in the opposite direction. If we know acceleration, we may want to recover velocity. If we know velocity, we may want to recover position:

$$\mathbf{a}(t) \quad \longrightarrow \quad \mathbf{v}(t) \quad \longrightarrow \quad \mathbf{r}(t).$$

This direction uses integration. Integration introduces constants or, equivalently, requires initial conditions. These are not mathematical noise; they represent missing information about the motion.

Theorem

If $\mathbf{v}(t) = d\mathbf{r}/dt$ on an interval containing $[t_0, t]$, then

$$\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^{t} \mathbf{v}(\tau)\, d\tau.$$

If $\mathbf{a}(t) = d\mathbf{v}/dt$ on an interval containing $[t_0, t]$, then

$$\mathbf{v}(t) = \mathbf{v}(t_0) + \int_{t_0}^{t} \mathbf{a}(\tau)\, d\tau.$$

The symbol τ is a dummy variable of integration. The upper limit t is the time at which we want to find the position or velocity. The lower limit t_0 is the initial time.

3.5.3 Why initial conditions are necessary

Knowing how something changes is not the same as knowing its value. Suppose we know only that

$$v(t) = 3.$$

Then

$$x(t) = \int 3\,dt = 3t + C.$$

The constant C is not determined by the velocity. It tells us the initial position. If $x(0) = 2$, then $C = 2$, so

$$x(t) = 3t + 2.$$

Example

Suppose acceleration is constant:

$$a(t) = A.$$

Then

$$v(t) = At + C_1.$$

If $v(0) = v_0$, then $C_1 = v_0$, so

$$v(t) = v_0 + At.$$

Integrating again gives

$$x(t) = v_0 t + \frac{1}{2}At^2 + C_2.$$

If $x(0) = x_0$, then $C_2 = x_0$, and therefore

$$x(t) = x_0 + v_0 t + \frac{1}{2}At^2.$$

This formula is often presented as a standard kinematic formula. Here it appears as the result of integrating constant acceleration twice and using initial conditions.

3.5.4 Vector form

The same idea works in vector form. If acceleration is known, then

$$\mathbf{v}(t) = \mathbf{v}(t_0) + \int_{t_0}^{t} \mathbf{a}(\tau)\,d\tau.$$

If velocity is known, then

$$\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^{t} \mathbf{v}(\tau)\,d\tau.$$

If acceleration is constant and equal to $\mathbf{A}$, and if $t_0 = 0$, then

$$\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{A}t,$$

and

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\mathbf{A}t^2.$$

3.5.5 The central structure of kinematics

The central structure of kinematics is

$$\boxed{\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}}$$

and

$$\boxed{\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}}.$$

The inverse relations are

$$\boxed{\mathbf{v}(t) = \mathbf{v}(t_0) + \int_{t_0}^{t} \mathbf{a}(\tau)\, d\tau}$$

and

$$\boxed{\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^{t} \mathbf{v}(\tau)\, d\tau}.$$

These formulas should not be treated as isolated recipes. They express the logic of kinematics: position describes where the body is, velocity describes how position changes, acceleration describes how velocity changes, and initial conditions supply the information needed when reconstructing motion by integration.

Dynamics will ask why acceleration appears. Before that question can be answered, we need the language developed in this chapter: position, velocity, acceleration, derivatives, integrals, and initial conditions.

Chapter 4

Dynamics

4.1 Introduction

This section provides an introduction to dynamics.

Chapter 5

Oscillations and Waves

5.1 Introduction

This section provides an introduction to oscillations and waves.

Chapter 6

Electromagnetism

6.1 Introduction

This section provides an introduction to electromagnetism.

Chapter 7

Statistics and Measurements

7.1 Introduction

This section provides an introduction to statistics and measurements.

Chapter 8

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8.1 Introduction

This section provides an introduction to cosmology.

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This section provides an introduction to relativity.

Chapter 10

Modern Physics

10.1 Introduction

This section provides an introduction to modern physics.

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